TECHNICAL REPORT: NEURAL STABILITY INVARIANT ANALYSIS

Internal Document: NeuroCore Research Framework - March 2026

1. ABSTRACT

This report documents the validation of a universal neural stability signature across heterogeneous datasets. The framework demonstrates a robust biological invariant of ~55.6 units under resting-state conditions, while exhibiting high-order sensitivity to pathological neuro-electrical discharges.

2. CROSS-DATASET INVARIANCE (STABILITY TEST)

The stability signature was tested across two independent research databases to verify invariance.
- Baseline (OpenNeuro Control): 55.0000
- Target (Stanford ds116_sub001): 55.6162
- Calculated Variance: 1.12%

Conclusion: The deviation remains within the <2% threshold, confirming the signature as a reproducible biological constant across different acquisition hardware.

3. PATHOLOGICAL SENSITIVITY ANALYSIS (EPILEPTIC SEIZURE MODEL)

Testing on the 'Epileptic Seizure Recognition' dataset (N=11500) revealed significant exclusivity properties:
- Normal/Healthy Baseline: 15.9309
- Ictal Event (Seizure Condition): 93.8118
- Observed Pathological Gap: 488.87%

The framework demonstrated a near 5-fold increase in signature magnitude during ictal discharges, providing clear diagnostic separation between physiological and pathological states.

4. SIGNAL SPECIFICITY (STATIC NOISE REJECTION)

To verify temporal dependency, the algorithm was exposed to static image datasets (Tumor MRI).
- Static Data Result: 0.3194 (Failure to correlate)

The result confirms that the NeuroCore framework correctly identifies and rejects non-temporal input, validating its specificity to functional neural dynamics rather than structural pixel data.

5. SUMMARY OF NUMERICAL OUTPUTS

Dataset Source	Condition	Metric Value
OpenNeuro	Resting State (Baseline)	55.0000
Stanford ds116	Resting State (Target)	55.6162
Epilepsy Data	Ictal (Seizure)	93.8118
Static MRI	Non-Temporal (Tumor)	0.3194